How effect sizes of invariant arcs shifted across the COVID break.
Green = strengthened, Red = weakened, Gray = stable.

Standardized Effect Size (σ)

0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

+0.64
+0.45
+0.39
+0.38
+0.36
+0.35
+0.27
+0.25
+0.24
+0.22
+0.20
+0.18
+0.17
+0.13
+0.05
+0.04
-0.15
-0.19
-0.31
-0.22
-0.35
-0.36
-0.26
-0.27
-0.28
-0.36
-0.38

+0.81 CrudeOil → INDPRO_chg
+0.66 CrudeOil → PCE_chg
+0.55 Russell3000 → PCE_chg
+0.54 SP500 → PCE_chg
+0.52 CrudeOil → CPI_chg
+0.49 Nasdaq → ConsumerConf_chg
+0.48 SP500 → ConsumerConf_chg
+0.42 Nasdaq → PCE_chg
+0.39 M2_chg → CrudeOil
+0.27 Russell3000 → VIX
+0.22 UNRATE_chg → PCE_chg
+0.17 FEDFUNDS_chg → INDPRO_chg
+0.01 CPI_chg → PCE_chg
-0.04 SP500 → M2_chg
-0.06 Russell3000 → M2_chg
-0.14 EuroUSD → CPI_chg
-0.15 INDPRO_chg → PCE_chg
0.21 REIT → M2_chg
-0.32 VIX → PCE_chg
-0.32 FEDFUNDS_chg → SP500
-0.36 FEDFUNDS_chg → Russell3000
-0.38 VIX → ConsumerConf_chg
-0.38 INDPRO_chg → CrudeOil
-0.44 PCE_chg → UNRATE_chg
-0.48 CrudeOil → M2_chg
-0.61 PCE_chg → M2_chg
-0.72 FEDFUNDS_chg → M2_chg

Pre-COVID
(Dec 2004 – Feb 2020)

Post-COVID
(Mar 2020 – Sep 2025)